A crystallographic restriction for multiplier cyclic sieving on necklaces

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Abstract

Let $Y_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q / [n]_q$ be the natural q-analogue of the number $\binom{n}{k}/n$ of binary necklaces of length n with k black beads, defined whenever $\gcd(n,k) = 1$. Rotation acts trivially on necklaces, but the *multipliers* $x \mapsto vx$ for $v \in (\mathbb{Z}/n\mathbb{Z})^\times$ do not; group-theoretically these are the symmetries lying in the normaliser of the rotation group inside S_n. We ask when $Y_{n,k}$ evaluated at a root of unity counts the necklaces a multiplier fixes.

Writing $f_e = \gcd(v^e - 1, n)$ for the number of points of $\mathbb{Z}/n\mathbb{Z}$ held fixed by v^e, we give a criterion: the sieving identity holds for every k coprime to n if and only if, for every proper divisor e of the order m of v, one has $\varphi(f_e) \le 2$ — that is $f_e \in \{1,2,3,4,6\}$ — together with $\mathrm{rad}(f_e) \mid m$. The first condition is the crystallographic restriction, and we show that it is forced: for multipliers with uniform orbits and $f < m$, the identity compels every prime $q \le f$ with $q \ne f-1$ to divide f itself, and elementary number theory then permits only $1, 2, 3, 4, 6$.

We prove an exact formula for the number of multiplier-fixed necklaces which makes the question computable, prove the arithmetic reduction to the crystallographic set, and report exhaustive verification of the criterion: 2,709 multipliers over $4 \le n \le 96$ with no counterexample, using exact integer and cyclotomic arithmetic throughout. The cases $f \in \{3,4,6\}$ lie in a regime that Stucky [2] explicitly leaves open.

1 Introduction

Looping a string makes a cyclic group act on it, and the characters of a cyclic group are roots of unity. The sharpest modern expression of that principle is the *cyclic sieving phenomenon* of Reiner, Stanton and White [1]; see [3] for a survey.

Definition 1.1 ([1])**.** Let X be a finite set carrying an action of a cyclic group $C = \langle c \rangle$ of order N, and let $X(q) \in \mathbb{Z}[q]$ satisfy $X(1) = |X|$. The triple $(X, X(q), C)$ exhibits the *cyclic sieving phenomenon* if for every $d \in \mathbb{Z}$,

$$\left|\{x \in X : c^d x = x\}\right| \;=\; X(q)\big|_{q=\zeta_N^d},$$

where $\zeta_N = e^{2\pi i/N}$.

*Discovered, verified and drafted in collaboration with Claude (Anthropic). All computations are exact and reproducible from the accompanying source code; see Section 7.

Fix $n \geq 3$ and colour the positions $\mathbb{Z}/n\mathbb{Z}$ with two colours, k of them black. Modulo the rotation action of $\mathbb{Z}/n\mathbb{Z}$ these are the *necklaces*; when $\gcd(n,k) = 1$ the rotation action is free and there are exactly $\binom{n}{k}/n$ of them. The natural q-analogue of that count is

$$Y_{n,k}(q) \;=\; \frac{\left[{n \atop k}\right]_q}{[n]_q}, \qquad [n]_q = 1 + q + \cdots + q^{n-1}, \tag{1}$$

which is a polynomial with non-negative integer coefficients precisely when $\gcd(n,k) = 1$ (Proposition 4.1).

Rotation itself acts trivially on necklaces, so it carries no information. What survives is the action of the *multiplier group* $(\mathbb{Z}/n\mathbb{Z})^\times$: for a unit v, relabel positions by $x \mapsto vx$. Inside S_n these are exactly the permutations normalising the cyclic rotation group, so the full symmetry available is the affine group $\mathbb{Z}/n\mathbb{Z} \rtimes (\mathbb{Z}/n\mathbb{Z})^\times$. Rotation spins a necklace; a multiplier rewires it.

Known results. For n prime the sieving always holds; this is Corollary 3.3 of Stucky [2], whose general theorem covers multipliers acting on $\mathbb{Z}/n\mathbb{Z}$ with exactly one or two fixed points, subject to conditions on the associated Young subgroup. Beyond that the paper stops, and says so:

> "*. . . when* $n \not\equiv 1, 2 \bmod m$ *the situation appears much more delicate, and we do not have a general conjecture.*" [2, §3]

That regime — composite n, multipliers with three or more fixed points — is the subject of this note.

Results. Section 3 proves an exact formula for the number of multiplier-fixed necklaces (Lemma 3.1); it is the tool that makes the question decidable in $O(n^2)$ arithmetic operations rather than by exponential enumeration. Section 4 records the cyclotomic factorisation of $Y_{n,k}$ and its evaluation at roots of unity. Section 5 states the criterion. Section 6 proves the arithmetic heart: the fixed-point count is forced into the crystallographic set $\{1, 2, 3, 4, 6\}$. Section 7 reports the computational evidence and Section 8 the boundaries. We are explicit throughout about which statements are proved and which are verified.

2 Notation

Throughout $n \geq 3$, and $\alpha = (\alpha_1, \ldots, \alpha_\ell)$ is a *content*: a composition of n recording how many positions receive each of ℓ colours. Words of content α are maps $w : \mathbb{Z}/n\mathbb{Z} \to \{1, \ldots, \ell\}$ with $|w^{-1}(i)| = \alpha_i$. The rotation group $\mathbb{Z}/n\mathbb{Z}$ acts on them, freely if and only if $\gcd(\alpha) = 1$; the orbits are *α-necklaces*. The binary case is $\alpha = (k, n-k)$ with $\gcd(n,k) = 1$.

For $v \in (\mathbb{Z}/n\mathbb{Z})^\times$ we write $m = \mathrm{ord}_n(v)$ and let v act on necklaces by $[w] \mapsto [w \circ v]$. We put

$$f_e = \gcd(v^e - 1,\ n) = \left|\{x \in \mathbb{Z}/n\mathbb{Z} : v^e x = x\}\right|, \qquad f = f_1.$$

Φ_d denotes the d-th cyclotomic polynomial, ζ_m a primitive m-th root of unity, φ Euler's totient, and $\mathrm{rad}(x) = \prod_{p \mid x} p$ the radical. We call v *uniform* if every $\langle v \rangle$-orbit on $\mathbb{Z}/n\mathbb{Z}$ has size 1 or m.

3 Counting multiplier-fixed necklaces

For $c \in \mathbb{Z}/n\mathbb{Z}$ let $A_c : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ be the affine map $A_c(y) = vy + c$; these are exactly the affine permutations with linear part v. Set

$$S \;=\; 1 + v + v^2 + \cdots + v^{m-1} \;\in\; \mathbb{Z}/n\mathbb{Z}, \qquad K \;=\; \{c \in \mathbb{Z}/n\mathbb{Z} : cS = 0\},$$

and let $N_\alpha(A_c)$ denote the number of words of content α that are constant on the cycles of A_c.

Lemma 3.1. *Let* $\gcd(\alpha) = 1$ *and* $v \in (\mathbb{Z}/n\mathbb{Z})^\times$. *The number of* α*-necklaces fixed by the multiplier* v *is*

$$\#\,\mathrm{Fix}(v) \;=\; \frac{1}{n} \sum_{c \in K} N_\alpha(A_c).$$

In particular every affine map A_c *with* $c \notin K$ *contributes nothing.*

Proof. Step 1 (which words count). A necklace $[w]$ is fixed by v iff $w \circ v = w \circ \rho_t$ for some rotation $\rho_t(x) = x + t$. Substituting $y = x + t$, this says $w \circ A = w$ for the affine map $A(y) = v(y - t)$, whose linear part is v. Conversely any w with $w \circ A_c = w$ has v-fixed necklace class. So $[w]$ is fixed iff w is constant on the cycles of some A_c.

Step 2 (no double counting). Suppose w is constant on the cycles of both A_c and $A_{c'}$ with $c \neq c'$. Then w is invariant under $A_c A_{c'}^{-1}$, which is the rotation by $c - c' \neq 0$. Since $\gcd(\alpha) = 1$ the rotation action on words of content α is free, a contradiction. Hence the sets $\{w : w \circ A_c = w\}$ are pairwise disjoint, and summing over c then dividing by the (free) rotation orbit size n gives the count, with no inclusion–exclusion.

Step 3 (only $c \in K$ *contribute).* Compute $A_c^m(y) = v^m y + c(1 + v + \cdots + v^{m-1}) = y + cS$, so A_c^m is the rotation by cS. Suppose $c \notin K$, i.e. $cS \neq 0$, and let $e > 1$ be its additive order. The rotation group generated by A_c^m has order e and acts freely on $\mathbb{Z}/n\mathbb{Z}$; since it is contained in $\langle A_c \rangle$, every $\langle A_c \rangle$-orbit has size divisible by e. A word constant on the cycles of A_c therefore has every colour class a union of cycles, so $e \mid \alpha_i$ for each i, whence $e \mid \gcd(\alpha) = 1$ — contradicting $e > 1$. Thus $N_\alpha(A_c) = 0$ for $c \notin K$. ☐

Remark 3.2. Step 3 is what makes the problem tractable: $|K| = \gcd(S, n)$ is typically a small divisor of n, so the sum has few terms, and each cycle type is computed in $O(n)$ steps.

4 The q-analogue at a root of unity

Proposition 4.1. *Let* $\gcd(n, k) = 1$. *Then* $Y_{n,k}(q)$ *of* (1) *is a polynomial, and*

$$Y_{n,k}(q) \;=\; \prod_{d \in E(n,k)} \Phi_d(q), \qquad E(n,k) = \{\, d \geq 2 \;:\; d \nmid n \;\; \textit{and} \;\; (k \bmod d) > (n \bmod d) \,\}.$$

Proof. From $[j]_q = \prod_{d \mid j,\, d \geq 2} \Phi_d(q)$ we get $[n]_q! = \prod_{d \geq 2} \Phi_d^{\lfloor n/d \rfloor}$, hence $\left[{n \atop k}\right]_q = \prod_{d \geq 2} \Phi_d^{\varepsilon_d}$ with $\varepsilon_d = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor$. Writing $a = k \bmod d$, $b = (n - k) \bmod d$, $c = n \bmod d$, we have $a + b \in \{c, c + d\}$ and $\varepsilon_d = (a + b - c)/d \in \{0, 1\}$, with $\varepsilon_d = 1$ iff $b = c - a + d$, i.e. iff $a > c$. Also $[n]_q = \prod_{d \mid n,\, d \geq 2} \Phi_d$. For $d \mid n$ with $d \geq 2$ we have $c = 0$ and, since $\gcd(n, k) = 1$ forces $d \nmid k$, also $a > 0 = c$; so $\varepsilon_d = 1$ and every factor of $[n]_q$ is cancelled exactly. The quotient is the stated product. ☐

Proposition 4.2 (q-Lucas)**.** *Let ζ be a primitive m-th root of unity, and write $n = am + r$, $k = bm + s$ with $0 \le r, s < m$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\zeta} = \binom{a}{b} \begin{bmatrix} r \\ s \end{bmatrix}_q \Big|_{q=\zeta}, \qquad [n]_q \Big|_{q=\zeta} = [r]_\zeta.$$

Consequently, if $m \nmid n$,

$$Y_{n,k}(\zeta) \;=\; \binom{a}{b} \cdot \frac{\left[\begin{smallmatrix} r \\ s \end{smallmatrix}\right]_q \big|_{q=\zeta}}{[r]_\zeta}. \tag{2}$$

Proof. The first identity is the q-Lucas theorem. The second holds because $\sum_{i=0}^{n-1} \zeta^i$ splits into a complete cycles, each summing to 0 when $m > 1$, leaving $\sum_{i=0}^{r-1} \zeta^i$. □

Corollary 4.3. *With $r \ge 1$ and $0 \le s$, the ratio in* (2) *satisfies*

$$\frac{\left[\begin{smallmatrix} r \\ s \end{smallmatrix}\right]_q \big|_{q=\zeta}}{[r]_\zeta} = \begin{cases} 1, & s \in \{1,\ r-1\}, \\ 0, & s > r, \\ 1/[r]_\zeta, & s \in \{0,\ r\}. \end{cases}$$

Proof. $\left[\begin{smallmatrix} r \\ 1 \end{smallmatrix}\right]_q = \left[\begin{smallmatrix} r \\ r-1 \end{smallmatrix}\right]_q = [r]_q$; $\left[\begin{smallmatrix} r \\ 0 \end{smallmatrix}\right]_q = \left[\begin{smallmatrix} r \\ r \end{smallmatrix}\right]_q = 1$; and $\left[\begin{smallmatrix} r \\ s \end{smallmatrix}\right]_q = 0$ for $s > r$. □

Thus (2) is manifestly a rational integer for $s \in \{1, r-1\}$ and vanishes for $s > r$. For the remaining residues it is an algebraic number in $\mathbb{Q}(\zeta)$ which, in every case we have examined, is irrational; since a fixed-point count is a rational integer, those residues are obstructions. We isolate this as a hypothesis rather than assert it.

Hypothesis 4.4. Let ζ be a primitive m-th root of unity and $2 \le r < m$. For $s \in \{0, 2, 3, \dots, r-2, r\}$ the quantity $\left[\begin{smallmatrix} r \\ s \end{smallmatrix}\right]_q \big|_{q=\zeta} / [r]_\zeta$ is not a rational number.

Hypothesis 4.4 has been verified for all $m \le 96$ and all admissible r, s in the course of the computations of Section 7; we have not proved it.

5 The criterion

Conjecture 5.1 (Criterion)**.** *Let $n \ge 3$ and $v \in (\mathbb{Z}/n\mathbb{Z})^\times$ of order $m > 1$, and put $f_e = \gcd(v^e - 1, n)$. Then*

$$Y_{n,k}(\zeta_m) \;=\; \#\{\, k\text{-subset necklaces of } \mathbb{Z}/n\mathbb{Z} \text{ fixed by } v \,\} \qquad \text{for every } k \text{ coprime to } n$$

if and only if, for every proper divisor e of m,

(i) $\varphi(f_e) \le 2$, *equivalently* $f_e \in \{1, 2, 3, 4, 6\}$; *and*

(ii) $\mathrm{rad}(f_e) \mid m$.

Since the cyclic sieving phenomenon of Definition 1.1 for the group $\langle u \rangle$ requires the identity at every power u^d, and u^d is itself a multiplier, Conjecture 5.1 determines the full CSP by conjunction over the divisors of $\mathrm{ord}_n(u)$.

Corollary 5.2 (uniform case)**.** *If v is uniform then $f_e = f$ for every proper divisor e of m, and the criterion reads: sieving holds if and only if $f \in \{1, 2, 3, 4, 6\}$ and $\mathrm{rad}(f) \mid m$. In particular the verdict depends only on the pair (m, f) — not on n, and not on the number of orbits.*

The independence from n asserted in Corollary 5.2 is a striking feature of the data and was the first indication that a clean statement existed; see Figure 1.

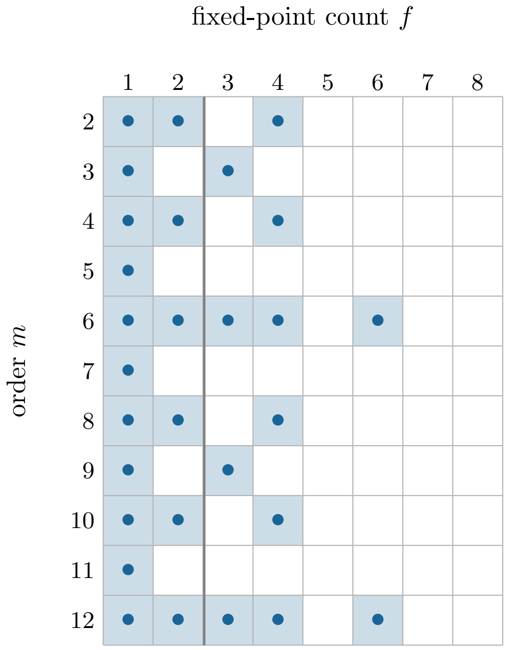

Figure 1: Uniform multipliers: sieving holds (•, shaded) or fails (blank), by the order m of the multiplier and its fixed-point count f. Columns $f = 5, 7, 8$ are empty for every m; the pattern of a row depends only on which primes divide m. Columns $f = 1, 2$ are the previously known cases, columns $f = 3, 4, 6$ are new.

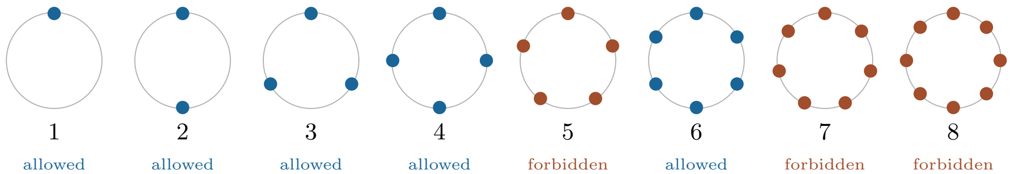

Figure 2: The permitted sizes of a multiplier's fixed set, $f \in \{1, 2, 3, 4, 6\}$ (Theorem 6.4) — identical to the permitted orders of rotational symmetry of a plane lattice, and for the same underlying reason: a quantity built from a root of unity is required to be rational.

6 The crystallographic restriction

The plane crystallographic restriction says that a lattice in $\mathbb{R}^2$ admits rotational symmetry only of order $1, 2, 3, 4$ or 6, because the trace $2\cos(2\pi/f)$ of the rotation must be a rational integer, i.e. $\varphi(f) \leq 2$. We now show that the same five numbers are forced here, by a different rationality constraint.

Lemma 6.1 (which residues occur)**.** *Let* $g = \gcd(m, n)$ *and* $0 \leq s < m$. *There exists* k *with* $k \equiv s \pmod{m}$ *and* $\gcd(k, n) = 1$ *if and only if* $\gcd(s, g) = 1$.

Proof. If a prime p divides both s and g then $p \mid n$ and $p \mid k$ for every $k \equiv s \pmod{m}$, so no such k is coprime to n. Conversely assume $\gcd(s, g) = 1$ and let $p \mid n$ be prime. If $p \mid m$ then $p \mid g$, so $p \nmid s$ and hence $p \nmid k$ automatically for every $k \equiv s \pmod{m}$. If $p \nmid m$ then the progression $s + m\mathbb{Z}$ meets every residue class modulo p, so we may avoid 0; the finitely many such conditions are simultaneously satisfiable by the Chinese remainder theorem. $\square$

Lemma 6.2 (uniformity constrains primes)**.** *Let v be uniform of order m, and let p be a prime with $p \mid n$ and $p \nmid f$. Then $\mathrm{ord}_p(v) = m$; in particular $m \mid p-1$ and $m \le p-1$.*

Proof. Let $H = (n/p)\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z}$, a $\langle v\rangle$-stable subgroup, and take $x = (n/p)u \in H$ with $p \nmid u$. Then $v^j x = x$ iff $n \mid (v^j - 1)(n/p)u$ iff $p \mid v^j - 1$, so the $\langle v\rangle$-orbit of x has size $\mathrm{ord}_p(v)$. If x were fixed then $p \mid v-1$ and $p \mid n$, so $p \mid \gcd(v-1, n) = f$, contrary to hypothesis. Hence x is not fixed, and uniformity forces its orbit size to be m, i.e. $\mathrm{ord}_p(v) = m$. As $\mathrm{ord}_p(v)$ divides $p-1$, the claim follows. $\square$

Lemma 6.3 (arithmetic)**.** *Let $f \ge 1$ be an integer such that every prime $q \le f$ with $q \ne f-1$ divides f. Then $f \in \{1,2,3,4,6\}$.*

Proof. The values $f = 1,2,3,4,6$ satisfy the condition: for $f = 6$ the primes ≤ 6 are $2,3,5$ and $5 = f-1$ is exempt, while $2 \mid 6$ and $3 \mid 6$; the others are immediate. For $f = 5$ the primes ≤ 5 are $2,3,5$ and $f-1 = 4$ is not prime, so $2 \mid 5$ would be required, which fails.

Now suppose $f \ge 7$. Every prime $q \le f/2$ satisfies $q \ne f-1$, because $f-1 > f/2$ for $f > 2$; hence every prime $q \le f/2$ divides f, and therefore

$$f \;\ge\; \prod_{p \le f/2} p \;=\; e^{\theta(f/2)},$$

where θ is Chebyshev's function. Iterating from $f \ge 7$: primes $2,3 \le f/2$ give $6 \mid f$, so $f \ge 12$; then $2,3,5 \le f/2$ give $30 \mid f$, so $f \ge 30$; then $2,3,5,7,11,13 \le f/2$ give $30030 \mid f$, so $f \ge 30030$. For $f \ge 30030$ the Chebyshev bound $\theta(x) > 0.84x$ (valid for $x \ge 101$) yields $f \ge e^{0.42f}$, which is false. Hence no $f \ge 7$ satisfies the condition. $\square$

Theorem 6.4 (necessity of the crystallographic restriction)**.** *Let v be a uniform multiplier of order m with $f < m$, and assume Hypothesis 4.4. If the sieving identity of Conjecture 5.1 holds for every k coprime to n, then*

$$f \in \{1,2,3,4,6\} \qquad \textit{and} \qquad \mathrm{rad}(f) \mid m.$$

Proof. Uniformity gives $n = f + mt$ where t is the number of orbits of size m, so $r := n \bmod m = f \bmod m = f$, using $f < m$. Let $s = k \bmod m$.

By (2), $Y_{n,k}(\zeta_m) = \binom{a}{b} \cdot \left[{r \atop s}\right]_q\big|_\zeta / [r]_\zeta$ with $a = \lfloor n/m \rfloor \ge b = \lfloor k/m \rfloor$, so $\binom{a}{b} \ge 1$. If s lies in the *bad set* $B = \{0, 2, 3, \ldots, r-2, r\}$ then, by Hypothesis 4.4, $Y_{n,k}(\zeta_m) \notin \mathbb{Q}$, whereas the right-hand side of the sieving identity is a non-negative integer. Hence no k coprime to n may have $k \bmod m \in B$. By Lemma 6.1 this says $\gcd(s, g) > 1$ for every $s \in B$, where $g = \gcd(m, n)$.

In particular, for every prime q with $2 \le q \le r = f$ and $q \ne f-1$ we have $q \in B$ and therefore $q \mid g$; so $q \mid m$ and $q \mid n$. Suppose such a q did not divide f. Then Lemma 6.2 applies to the prime q and gives $m \le q-1$; but $q \mid m$ gives $q \le m$, so $q \le m \le q-1$, a contradiction. Hence every prime $q \le f$ with $q \ne f-1$ divides f, and Lemma 6.3 yields $f \in \{1,2,3,4,6\}$.

Finally we check $\mathrm{rad}(f) \mid m$. Let q be any prime dividing f. Then $q \ne f-1$: otherwise q would divide both f and $f-1$, hence $q \mid 1$. So q falls under the case already treated, giving $q \mid g \mid m$. As this holds for every prime divisor of f, we conclude $\mathrm{rad}(f) \mid m$. $\square$

Remark 6.5. Theorem 6.4 covers uniform multipliers with $f < m$. When $f \geq m$ (in particular when $m \mid n$, where $r = 0$ and Proposition 4.2 degenerates) the same conclusion is observed in every computed case but the argument above does not apply verbatim. Sufficiency, and the multi-level form of Conjecture 5.1 for non-uniform v, remain unproved.

Remark 6.6 (why the coincidence is not one). Both the lattice statement and Theorem 6.4 amount to demanding that a quantity built from a root of unity be rational: for a lattice it is the trace of a rotation, here it is a Gaussian binomial divided by a q-integer. In each case the demand confines the root of unity to a field of degree at most two over $\mathbb{Q}$, and the surviving orders are $\{1, 2, 3, 4, 6\}$. A necklace contains no lattice; what the two situations share is only the circle.

7 Computational verification

All computations use exact integer arithmetic, together with exact arithmetic in $\mathbb{Z}[q]/(\Phi_m(q))$ for root-of-unity evaluations. No floating point is used anywhere.

Method. The polynomial side is computed from Proposition 4.1 as a product of cyclotomic polynomials reduced modulo Φ_m; the fixed-point side from Lemma 3.1. Both were cross-validated against independent routes: the factorisation against direct polynomial division of $\begin{bmatrix} n \\ k \end{bmatrix}_q$ by $[n]_q$ for all $n < 30$, and the fixed-point formula against brute-force enumeration of necklaces.

Scope. Each multiplier is tested against *every* admissible k, so the number of individual instances is in the tens of thousands.

Form of the criterion	range	multipliers	counterex.
General (all $e \mid m$, no uniformity)	$4 \leq n \leq 96$	2,709	0
Uniform (m, f) form (Cor. 5.2)	$4 \leq n \leq 115$	2,682	0

Table 1: Exhaustive verification of Conjecture 5.1.

8 Boundaries and open problems

Uniformity is not necessary. Of 280 non-uniform multipliers with $n \leq 60$, exactly 38 satisfy the sieving identity; all of them have orbit-size profile $\{1, 2, m\}$. This is why Conjecture 5.1 is stated over all divisors $e \mid m$ rather than in the (m, f) form: the single-level condition on $f = f_1$ is necessary but not sufficient, failing in 106 of 603 multipliers with $n \leq 46$.

Three colours is strictly stronger. For contents with $\ell \geq 3$ colours the criterion as stated is no longer correct: among 213 multipliers with $n \leq 28$ there are 25 for which the two-colour identity holds but the three-colour identity fails — the smallest at $n = 28$. In every discrepancy the criterion over-predicts. The additional obstruction is not understood.

Open problems.

1. Prove Hypothesis 4.4.
2. Prove sufficiency in Conjecture 5.1, and extend Theorem 6.4 to $f \geq m$.
3. Determine the correct criterion for contents with $\ell \geq 3$ colours.

A An unrelated side result: orbit towers terminate

A sequence $a = (a_j)_{j\geq 1}$ of non-negative integers is *realisable* if $L(a)_j = \frac{1}{j}\sum_{d|j}\mu(j/d)\,a_d$ is again a sequence of non-negative integers; then a_j counts the points of period j of a dynamical system and $L(a)_j$ counts its orbits of length j — its "primes". One may ask whether the primes of a system can themselves be the points of a system, indefinitely.

Theorem A.1. *Let $a \neq 0$ be a sequence of non-negative integers. Then $L^j(a)$ fails to be a sequence of non-negative integers for some $j \geq 1$.*

Proof. Let $N = \min\{j : a_j \neq 0\}$. If $N \geq 2$ then $a_j = 0$ for $j < N$, and this persists: if $c_j = 0$ for all $j < N$ then $L(c)_j = 0$ for $j < N$, while $L(c)_N = c_N/N$ since every proper divisor of N contributes 0. Hence $L^j(a)_N = a_N/N^j$, which is a positive integer for all j only if $a_N = 0$, a contradiction.

If $N = 1$, put $\lambda = a_1 > 0$ and $x_j = L^j(a)_2$. Since L fixes the first term, $L^j(a)_1 = \lambda$ for all j, and $x_{j+1} = (x_j - \lambda)/2$. Solving, $x_j = -\lambda + (a_2 + \lambda)2^{-j} \to -\lambda < 0$, so $x_j < 0$ for large j. ☐

Thus the tower of "primes of the primes of the primes" is always finite, with height at most $\log_N(a_N)$ when $N \geq 2$. Note the asymmetry: the opposite direction, $a \mapsto (\sum_{d|j} d\,a_d)_j$, may be iterated forever.

B Attribution ledger

The results above emerged from a broad computational search, most of whose output was already known. Recording that is part of the result.

- **New.** Conjecture 5.1 and Theorem 6.4; the cases $f \in \{3, 4, 6\}$ are not covered by [2], and lie exactly in the regime that paper flags as having no general conjecture.
- **New.** Lemma 3.1, in particular the vanishing for $c \notin K$.
- **Rediscovered.** Sieving for prime n: derived and proved here via q-Lucas before being located as [2, Cor. 3.3]. The result is Stucky's.
- **Rediscovered.** That $([n]_q/[n-mk]_q)\left[{n-mk \atop k}\right]_q$ cyclically sieves the k-subsets of a cycle with all cyclic gaps $\geq m+1$ (verified to $n = 16$).
- **Rediscovered.** That the central binomial, Apéry, Franel and Domb numbers are periodic-point counts; their orbit counts are catalogued in the OEIS as the `A060165` family.

- **Observed, uncatalogued.** Necklaces equal to their own Burrows–Wheeler transform. For $n = 2, \ldots, 18$ their numbers are

$$3, 4, 4, 4, 7, 6, 5, 9, 5, 6, 9, 8, 8, 11, 11, 7, 15,$$

so among the 4116 necklaces of length 16 exactly 11 are BWT-fixed. We found no database entry for this sequence and offer no characterisation.

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